Product Requirements Document (PRD)

Restaurant Payment Splitting Application

1. Executive Summary

Product Name: SplitCheck (working title)

Product Vision: Streamline the restaurant payment process by enabling customers to seamlessly split bills through their mobile devices, eliminating manual calculation and reducing payment friction.

Problem Statement: Currently, restaurant customers must manually inspect physical receipts to determine individual payment amounts, leading to confusion, calculation errors, and delays in the payment process. This creates friction for both customers and restaurant staff.

2. Product Overview

2.1 Core Value Proposition

- **For Customers:** Effortless bill splitting with clear visibility of individual orders and automated payment processing
- **For Restaurants:** Reduced payment processing time, improved customer satisfaction, and streamlined operations

2.2 Target Users

- **Primary:** Restaurant customers dining in groups
- **Secondary:** Restaurant waitstaff and management
- **Tertiary:** Restaurant owners and POS system administrators

3. Current State Analysis

3.1 Current Customer Journey (Steps 1-12)

1. Group enters restaurant and sits down
2. Menu access (physical or QR code)
3. Waitress approaches for order taking
4. Order recorded on paper
5. Order communicated to kitchen/bar staff
6. Food preparation and delivery cycle

7. Repeat ordering as needed
8. Check request initiated
9. **Pain Point**: Waitress manually enters order into POS system
10. **Pain Point**: Payment method discussion
11. **Major Pain Point**: Manual bill splitting and calculation
12. Individual payment processing

3.2 Identified Problems

- Manual bill inspection and calculation (Step 11)
- Potential for calculation errors
- Time-consuming payment resolution
- Customer confusion over individual amounts
- Risk of duplicate or missed payments

4. Proposed Solution

4.1 Product Integration Point

The application integrates at **Step 9** of the current process, transforming the payment experience from Steps 9-12.

4.2 Core Functionality

Enhanced Payment Flow:

1. **POS Integration**: When waitress enters order details, system automatically updates customer-facing interface
2. **QR Code Activation**: Existing table QR codes become gateways to payment sessions
3. **Live Session Management**: Customers join shared payment room with real-time visibility
4. **Interactive Bill Splitting**: Customers select their items with immediate visual feedback
5. **Integrated Payment Processing**: Direct bank account or card payments within the app
6. **Staff Notifications**: Real-time updates to waitstaff on payment progress

5. Functional Requirements

5.1 Core Features

5.1.1 Customer-Facing Features

- **QR Code Scanning**: Access payment session via table QR code
- **Live Session Room**:
  - View active participants count
  - See real-time item selection status
  - Display total bill breakdown
- **Item Selection Interface**:
  - Browse ordered dishes with quantities and prices
  - Select personal items with one-click
  - View running total of selected items
- **Payment Processing**:
  - Connected bank account/card integration
  - Custom tip calculation and selection
  - Secure payment confirmation
- **Visual Feedback**:
  - Locked/disabled items when selected by others
  - Clear indication of personal total
  - Payment status visibility

5.1.2 Staff-Facing Features

- **POS Integration**: Seamless data sync from existing POS systems
- **Mobile Dashboard**:
  - View active payment sessions
  - Monitor payment completion status
  - Access order details and history
- **Notification System**:
  - Push notifications for completed payments
  - Alert system for session issues
- **Session Management**:
  - Ability to reset QR codes for new customers
  - Override capabilities for dispute resolution

5.1.3 Administrative Features

- **Order History**: Complete transaction records with unique IDs
- **Analytics Dashboard**: Payment completion rates, session duration metrics
- **Staff Management**: Waitress ID tracking and performance metrics
- **QR Code Management**: Generate and reset table-specific codes

5.2 Technical Requirements

5.2.1 System Architecture

- **Real-time Communication**: WebSocket implementation for live session updates
- **POS Integration**: API connections to existing POS systems
- **Payment Gateway**: Secure integration with banking/payment providers
- **Database**: Scalable storage for orders, sessions, and user data
- **Mobile Optimization**: Responsive web application or native mobile app

5.2.2 Security Requirements

- **Data Encryption**: End-to-end encryption for payment data
- **PCI Compliance**: Meet payment card industry standards
- **Session Security**: Unique session tokens and automatic expiration
- **Bank Integration**: Secure API connections with financial institutions

5.2.3 Performance Requirements

- **Real-time Updates**: < 1 second latency for item selection updates
- **Payment Processing**: < 10 seconds for payment confirmation
- **Session Capacity**: Support up to 10 customers per session
- **Uptime**: 99.9% availability during business hours

6. User Stories

6.1 Customer Stories

- **As a customer**, I want to scan a QR code to access the bill so that I can see what was ordered
- **As a customer**, I want to select only the items I ordered so that I pay the correct amount
- **As a customer**, I want to see when others select items so that I avoid duplicate payments
- **As a customer**, I want to add a custom tip amount so that I can show appreciation for service
- **As a customer**, I want to pay directly through the app so that I don't need to handle cash or cards

6.2 Staff Stories

- **As a waitress**, I want the system to automatically update when I enter orders so that customers can immediately access their bill
- **As a waitress**, I want to receive notifications when payment is complete so that I know when customers can leave
- **As a waitress**, I want to view payment status on my phone so that I can assist customers if needed

7. User Experience Flow

7.1 Happy Path Scenario

1. Waitress enters complete order into POS system
2. System activates QR code for the table
3. Waitress instructs customers to scan QR code
4. Customers scan and join payment session
5. Customers see live participant count and full bill
6. Each customer selects their ordered items
7. Selected items become locked for other participants
8. Customers add tip percentage to their subtotal
9. Customers complete payment via connected bank account
10. System confirms individual payments
11. Waitress receives completion notification
12. All payments confirmed, session closes
13. QR code resets for next customers

7.2 Edge Cases

- **Disputed Items**: Override mechanism for waitstaff
- **Payment Failures**: Retry mechanisms and alternative payment methods
- **Partial Groups**: Support for customers who don't participate digitally
- **Session Abandonment**: Automatic timeout and manual reset options

8. Success Metrics

8.1 Primary KPIs

- **Payment Completion Time**: Reduce average time from bill presentation to full payment by 60%

- **Customer Satisfaction**: Achieve 4.5+ rating for payment experience
- **Adoption Rate**: 80% of eligible tables use the system within 6 months

8.2 Secondary Metrics

- **Error Reduction**: 95% reduction in payment calculation disputes
- **Staff Efficiency**: 30% reduction in payment-related staff time
- **Session Completion Rate**: 90% of started sessions result in full payment

9. Technical Considerations

9.1 POS System Integration

- **Compatibility Requirements**: Support for major POS systems (Square, Toast, Clover)
- **Data Mapping**: Standardized item identification across different POS formats
- **Real-time Sync**: Immediate data transmission upon order entry

9.2 Mobile Optimization

- **Cross-Platform**: Web-based solution compatible with iOS and Android
- **Responsive Design**: Optimized for various screen sizes
- **Offline Capability**: Basic functionality during connectivity issues

10. Implementation Phases

10.1 Phase 1: MVP (Months 1-3)

- Basic QR scanning and session creation
- Simple item selection interface
- Mock payment processing (no real transactions)
- Single POS integration pilot

10.2 Phase 2: Core Features (Months 4-6)

- Real payment processing integration
- Enhanced UI/UX with real-time updates
- Staff notification system
- Multiple POS system support

10.3 Phase 3: Advanced Features (Months 7-9)

- Analytics dashboard
- Advanced tip customization
- Customer payment history
- Multi-restaurant deployment

11. Assumptions and Dependencies

11.1 Assumptions

- Restaurants have existing QR code infrastructure or willingness to implement
- Customers are comfortable using mobile devices for payments
- POS systems provide accessible APIs for integration
- Payment processors support the required integration methods

11.2 Dependencies

- POS system partnership agreements
- Payment gateway partnerships
- Banking integration certifications
- Restaurant pilot program participation

12. Risk Assessment

12.1 Technical Risks

- **POS Integration Complexity**: Varying API standards across systems
- **Payment Security**: Compliance with financial regulations
- **Real-time Performance**: Scalability under high concurrent usage

12.2 Business Risks

- **Customer Adoption**: Resistance to new payment methods
- **Restaurant Buy-in**: Reluctance to change established processes
- **Competitive Response**: Similar solutions entering market

12.3 Mitigation Strategies

- Pilot program with select restaurants to validate concept
- Comprehensive security audit and compliance review
- Staff training programs to encourage adoption

- Flexible integration approach to accommodate various POS systems

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